MUHAMMAD SHUJA ALI

Systems Architect & Lead Embedded / Full-Stack Engineer

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TECHNICAL SKILLS

Languages & Firmware:	C, C++, Python, TypeScript, JavaScript, SQL, HTML5 / CSS3
Embedded & Hardware:	FreeRTOS, ESP-IDF, ESP32, STM32, Arduino, MicroPython, Modbus RTU (RS-485), ESP-NOW, I2C, SPI, UART, Energy Metering Telemetry, Digital MEMS Vibration Sensors, Embedded Board Programming
Frontend:	React, Next.js, Tailwind CSS, Chakra UI, Ant Design, WebSockets, High-Density Data Telemetry Dashboards
Backend, DB & Cloud:	Node.js, Express, Fastify, Flask, Firebase (Auth, RTDB, Functions), TimescaleDB, InfluxDB, PostgreSQL, Redis, GCP (Compute Engine, Pub/Sub), Docker, Nginx, CI/CD, Termux

PROFILE

Systems Architect and Full-Stack IoT Engineer with extensive experience programming embedded boards, developing deterministic firmware, and architecting scalable cloud telemetry platforms. Lead Engineer at KunjiSoft, where I built and deployed edge-to-cloud telematics platforms powering continuous industrial monitoring across 300+ heavy manufacturing machines in tier-1 manufacturing facilities. Strong expertise in deterministic firmware design (FreeRTOS, Modbus RTU, SPI/I2C), high-throughput time-series databases, and real-time operational interfaces.

EMPLOYMENT HISTORY

Director — KunjiSoft

May 2020 – Present

industrialpmr (IPMR226) Telematics Platform

- Designed and implemented an end-to-end edge telematics platform—encompassing embedded firmware programming, sensor integration, scalable backend services, and interactive web dashboards—operational across 300+ heavy manufacturing machines.
- Programmed embedded board firmware to interface with high-accuracy energy metering and digital vibration sensors for precise power and operational telemetry.
- Developed deterministic FreeRTOS firmware with preemptive task scheduling and a non-volatile SPI ring buffer to guarantee zero data loss across 72+ hours of factory network downtime.
- Implemented multi-drop Modbus RTU (RS-485) communication pipelines polling real-time power, vibration, and duty cycle parameters across distributed industrial plant equipment.
- Deployed and field-validated hardware and communications infrastructure at tier-1 manufacturing facilities, ensuring continuous operation in high-EMI environments under 40°C+ ambient temperatures, securing enterprise procurement.
- Constructed real-time WebSocket ingestion pipelines and high-density telemetry dashboards with sub-second plant floor status updates.

Developer — ShopsQual

Jun 2019 – May 2020

- Built performant web application components using ReactJS, Python Flask, PHP, HTML/CSS, and JavaScript.
- Implemented responsive client interfaces and integrated RESTful backend APIs for multi-tenant e-commerce workflows.

EMPLOYMENT HISTORY (CONTINUED)

Web Manager — Kunji Industrial Solutions — Apr 2018 – Jun 2019

- Maintained enterprise web properties, product catalogs, and client portal interfaces using HTML, CSS, and JavaScript.
- Improved page performance, cross-browser compatibility, and search visibility across company web assets.

EDUCATION

Bachelor of Science in Computer Science (BSCS) — Graduated
Iqra University, Karachi

ACHIEVEMENTS & AWARDS

2nd Prize Winner — AKU SIMVENTION Expo 2025 — Sep 2025
Awarded 2nd Prize at the Aga Khan University (AKU) SIMVENTION Expo for the **Smart Doser**—an automated, precision clinical medical dosing device featuring closed-loop motor actuation, safety interlocks, and dosage telemetry.

CERTIFICATIONS

Frontend Developer (React) Certificate — HackerRank	Jul 2024 · Credential ID: 5a31e577766a
React — HackerRank	Jan 2022 · Credential ID: a5a9d70376e1
JavaScript (Intermediate) — HackerRank	Jan 2022 · Credential ID: be96a7f07dad
JavaScript — HackerRank	Jan 2022 · Credential ID: 1cc8732a169e
Python — HackerRank	Jan 2022 · Credential ID: 619e04e75239
Node — HackerRank	Jan 2022 · Credential ID: 2cd327c54469
CSS — HackerRank	Jan 2022 · Credential ID: 8b98bfc62805

REPRESENTATIVE PORTFOLIO PROJECTS

Delivery Scheduler Simulation Engine — Live Demo · GitHub

- Logistics simulation system engineered to solve multi-vehicle package dispatch challenges under delivery deadlines and vehicle volume limits using dynamic programming and greedy heuristic algorithms.

MedQueue — Clinic Management & Queue Engine — GitHub

- High-concurrency outpatient clinic orchestration platform featuring a deterministic queue state machine, sub-50ms WebSocket broadcast pipeline, and automated turn-based patient dispatch.